Hazael Soledade de Araujo Jumonji

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Education

Shanghai Jiao Tong University (SJTU) China, 09/2025–03/2027
M.Sc. in Electronic and Information Engineering (SJTU Paris Elite Institute) - GPA: 3.89/4.0.
- Double degree with ENSTA Paris.
- Thesis: *Blockchain-based Trust Architecture for AI Agents Communicating via MCP and A2A*.
- Advisor: Prof. Yue Wu.

ENSTA Paris - Institut Polytechnique de Paris France, 08/2022–03/2027
M.Sc. Equivalent - Diplôme d'Ingénieur (*grade de master*), expected 03/2027 through the SJTU double degree. Diplôme de gradué en ingénierie (B.Sc.-equivalent), 2025.
- Specialization: Artificial intelligence, big data and systems engineering.

University of São Paulo (USP) Brazil, 02/2019–12/2021
Computer Engineering (undergraduate) - Weighted average: 8.8/10. Three years, then transfer to ENSTA Paris.

Research Experience

Shanghai Jiao Tong University - Prof. Yue Wu's group, School of Computer Science China, 09/2025–Present
Master's thesis researcher; research intern in the group, 08/2026–11/2026.
Project: **CrossTrust** - a trust layer with no single trusted party for AI agents of different organizations (MCP, A2A).
- Designed and built the full system: smart contracts with quorum governance on a permissioned Byzantine-fault-tolerant ledger (Hyperledger Besu, Solidity), a Python trust module that plugs into MCP and A2A, multi-agent testbeds and a benchmarking pipeline.
- Security: compared with **18 other systems** run from their public code, it stops **all 7 identity attacks** on both protocols (none of the others stops more than 5) and every tool swap by another party, even on first contact. With a real LLM agent, a swapped poisoned tool is followed in 12% of cases without a defense and in **none** with CrossTrust.
- Reputation: fake identities do not help; faking an agent's reputation takes **4 independent operators**, while in a public on-chain registry one operator's own fake identities suffice.
- Proofs: reputation guarantees machine-checked in **Lean 4**; the signed field each call carries verified in **Tamarin**.
- Overhead: under 3 ms of checks per call (0.2% of one LLM call); one receiver handles 400 calls per second while 99% of calls still finish within 1 s.

LIX - CNRS & École Polytechnique (MAX team) France, 06/2025–08/2025
Research intern, supervised by Joris van der Hoeven (CNRS research director).
Project: fast integer-matrix multiplication with Intel AMX.
- Wrote INT8 and INT16 kernels for arbitrary sizes, reaching up to **1,784 GOPS** (INT8, Xeon Platinum 8480+), with 98% of the cycles in the multiplication itself on large matrices.

Manuscripts

[1] **Jumonji, H. S. A.**, Wu, Y., & Hong, J. (2026). CrossTrust: A Blockchain-Based Trust Architecture for Cross-Organization AI Agents over MCP and A2A. *Manuscript in preparation.*

Grants and Awards

International Mathematics Competition for University Students (IMC) Online (UCL), 2021
Third Prize in the 28th edition; participant in 2020.

China Scholarship Council (CSC) - Chinese Government Scholarship China, 09/2025–03/2027
Merit-based scholarship covering all costs and a monthly stipend at SJTU; renewed competitively.

Elon Lages Lima University Mathematics Competition (OBM) Brazil, 2021
Bronze Medal in the 1st edition.

Ibero-American University Mathematics Competition (CIIM) Online, 2020
Representative of the University of São Paulo.

Online Physics Olympiad - Peruvian Society of Physics Teachers International, 2015–2016
1st Place (individual) in 2016 and 2nd Place in 2015, with students from Peru, Brazil, Mexico, Spain (and Panama in 2015).

Brazilian National Olympiads Brazil, 2014–2016
Gold Medals in Physics (2015) and Astronomy (2015, 2016); Silver Medal in Physics (2014); national Bronze Medal in Robotics, theoretical modality (2016).

Selections for International Olympiads Brazil, 2015–2016
Invited to the Brazilian selections for the IPhO (2016) and the IOAA (2015, 2016); Certificate of Honor from the Legislative Assembly of the State of Ceará for olympiad results (2016).

Need-based aid France and Brazil, 2019–2026
ENSTA Paris scholarship, partial (2022–2023) then full (2023–2026); financial aid at the University of São Paulo (2019–2022).

Merit scholarships Brazil, 2014–2018
Full merit scholarships at Colégio Helyos (2014), Colégio Farias Brito (2014–2017) and Curso Poliedro (2018).

Teaching Experience

NEMO-USP Math Club, ICMC - University of São Paulo Brazil, 01/2022–08/2022
Co-founder and coordinator: trained 15 students in algebra, combinatorics and geometry; organized a math competition at ICMC; members won 10+ international awards.

Volunteer Mentor - University of São Paulo Brazil, 2019–2021
Free tutoring for undergraduates in core engineering courses.

Volunteer Olympiad Instructor - University of São Paulo Brazil, 2019–2020
Free preparation classes in mathematics, physics and chemistry for students aiming at the Brazilian olympiads.

Other Experience and Activities

Autonomous Race Car Team - ENSTA Paris France, 09/2024–04/2025
2nd place in the Paris-Saclay autonomous car race (CoVAPSy 2025). Developed the simulator, the camera module, a YOLO network for obstacle and track detection on a Raspberry Pi, and LiDAR-based collision avoidance.

Member - ZENITH Aerospace (University of São Paulo) Brazil, 03/2020–07/2022
Student space group for stratospheric probes and CubeSats. Built the physical-mathematical model of a rocket launched from a stratospheric balloon (Project ZEUS, with the rocketry group Topus).

Software Development Intern - University of São Paulo Brazil, 10/2021–09/2022
Full-stack web platform for managing academic events (700-hour project); database design and SQL backend optimization.

Member - GANESH Information Security Group (University of São Paulo) Brazil, 02/2019–07/2019

Volunteer Work

Cop1 - Solidarités Étudiantes France, 2025
40 hours of volunteer work distributing food to students in need.

Languages

Language	Proficiency level
Portuguese	Native
English	Fluent
French	Fluent
Chinese	A2

Skills

Programming languages	Systems and security	Agents, AI and formal methods
Python (asyncio), Solidity, C/C++, SQL, JavaScript	Hyperledger Besu (QBFT), smart contracts, applied cryptography (Ed25519, SHA-256, TLS), Intel AMX, HPC clusters	Model Context Protocol (MCP), Agent2Agent (A2A) SDK, multi-agent systems, local LLMs, YOLO, Lean 4, Tamarin